

LIMPOPO POULTRY HATCHERY-PRODUCTION ANALYSIS

Small-Business Production Analysis Study | Limpopo, South Africa

Analysis Period: 7 January to 27 May 2022.

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Excel Data Analytics Portfolio Project

Limpopo Poultry Hatchery — Production Analysis

Introduction

This report presents a data quality review and production analysis of weekly hatchery records from six poultry farms across Limpopo province, covering the period 7 January to 27 May 2022. The purpose of the exercise was to take raw, weekly production logs and turn them into a clean, analysis-ready dataset, then use that dataset to identify performance trends, data quality issues, and operational recommendations across the farm network.

The workbook was built and analysed in Excel: Power Query-style manual cleaning was used to correct and standardise the raw data, pivot tables summarise hatchability and fertility by farm and quarter, and a single-page dashboard together with a written analysis sheet present the key findings and recommendations below.

Data Overview

The analysis is based on a single weekly production log covering all six farms, supplemented by a documented cleaning process. The workbook contains six tabs, each capturing a distinct part of the process:

1. Raw Data (Limpopo_poultryhatchery_messy)

Weekly hatchery records exactly as logged, 127 rows across 23 columns. Key columns include:

- **Week_Ending** — date the weekly record was closed off
- **Farm_Name / Farm_Type / District / Municipality** — identifies which of the 6 farms the row belongs to
- **Flock_ID** — unique flock identifier per farm (FL-001 to FL-006)
- **Eggs_Set, Eggs_Fertile, Eggs_Hatched** — weekly incubation volumes
- **Chicks_Graded_1st/2nd, Chicks_Culled, Mortalities** — chick outcome data
- **Biosecurity_Check, Vaccine_Given, Notes** — operational and health record fields

2. Data Quality Report

A row-by-row log of every issue identified in the raw data, the number of records affected, and how each was resolved (see Data Cleaning section below for the full breakdown).

3. Cleaned Data (Limpopo_poultryhatchery_clean)

The standardised, corrected dataset used for all analysis, pivots, and the dashboard. Adds calculated fields — Month, Quarter, Cull Rate %, and QC flags — on top of the raw columns.

4. Pivot Tables

Average Hatchability % and Fertility % by farm and by quarter, with minimum and maximum values.

5. Analysis

Key performance indicators, a per-farm performance summary table, and 8 written insights and recommendations.

6. Dashboard

A single-page visual summary of production volumes, hatchability trends, and mortality by farm (see below).

Recommended Analysis

- How much production did each farm generate this period, and how does hatchability compare across farms?
- Are there any farms or flocks that should be flagged for closer review?
- Which weeks or events caused production gaps, and how were they handled?
- How many records fell below the acceptable hatchability threshold?
- Which farms are least feed-efficient per chick hatched?
- What is the relationship between fertility rate and chick viability (cull rate, mortality)?

Data Cleaning

To ensure accuracy and consistency, the raw data was reviewed line by line and each issue logged in the Data_Quality_Report tab before being resolved in the cleaned dataset. The issues identified and how they were resolved are summarised below.

Issue	Records	Resolution
Inconsistent date formats	3	Normalised to yyyy-mm-dd
Breed name inconsistencies	2	Title-cased (Ross308 → Ross 308, lohmann brown → Lohmann Brown)
Farm Type / District case issues	2	Title-case standardisation applied
Biosecurity FAIL records	3	Retained, flagged for review (QC_Flag = Biosec Fail)
Shutdown / zero-production weeks	6	Excluded from clean data; retained in raw data
Duplicate row (FL-001, 2022-04-29)	1	Removed — one record kept per Farm + Week
Eggs Set outlier (999,999)	1	Removed (a correction to ~25,000 was possible but not applied)
Low hatchability (<91%)	30	Retained, flagged for management review

Summary: Raw rows = 127 | Rows removed/excluded = 9 | Cleaned rows = 118 (documented) | Data completeness after cleaning = 92.9%

Pivot/Analysis

PIVOT 1 — Avg Hatchability % by Farm & Quarter					
Farm_Name	Farm_Type	Quarter	Hatchability		
Min	Max				
☐ Bela-Bela Agri	☐ Broiler	Q1	90,90	90,09	91,43
		Q2	91,32	90,67	91,95
	Broiler Total		91,06	90,09	91,95
Bela-Bela Agri Total			91,06	90,09	91,95
☐ Giyani Poultry	☐ Broiler	Q1	91,51	91,11	91,97
		Q2	91,65	90,80	91,97
	Broiler Total		91,56	90,80	91,97
Giyani Poultry Total			91,56	90,80	91,97
☐ Lephalale Poultry Farm	☐ Broiler	Q1	91,70	91,39	92,33
		Q2	91,86	91,39	92,33
	Broiler Total		91,77	91,39	92,33
Lephalale Poultry Farm Total			91,77	91,39	92,33
☐ Mokopane Layers	☐ Layer	Q1	90,46	89,61	90,97
		Q2	90,57	89,96	90,97
	Layer Total		90,50	89,61	90,97
Mokopane Layers Total			90,50	89,61	90,97
☐ Polokwane Hatchery Hub	☐ Broiler	Q1	91,74	91,18	92,08
		Q2	91,67	90,75	92,08
	Broiler Total		91,72	90,75	92,08
Polokwane Hatchery Hub Total			91,72	90,75	92,08
☐ Tzaneen Breeder Farm	☐ Breeder	Q1	91,86	91,31	92,11
		Q2	91,86	91,40	92,11
	Breeder Total		91,86	91,31	92,11
Tzaneen Breeder Farm Total			91,86	91,31	92,11
Grand Total			91,41	89,61	92,33

PIVOT 2— Avg Fertility % by Farm & Quarter

Farm_Name	Farm_Type	Quarter	Fertility %	Min	Max
Bela-Bela Agri	Broiler	Q1	94,25	91,00	97,00
		Q2	95,00	93,00	97,00
	Broiler Total		94,55	91,00	97,00
Bela-Bela Agri Total			94,55	91,00	97,00
Giyani Poultry	Broiler	Q1	94,27	91,00	97,00
		Q2	94,25	91,00	97,00
	Broiler Total		94,26	91,00	97,00
Giyani Poultry Total			94,26	91,00	97,00
Lephalale Poultry Farm	Broiler	Q1	95,08	93,00	96,00
		Q2	95,25	94,00	96,00
	Broiler Total		95,15	93,00	96,00
Lephalale Poultry Farm Total			95,15	93,00	96,00
Mokopane Layers	Layer	Q1	94,25	92,00	96,00
		Q2	94,63	93,00	96,00
	Layer Total		94,40	92,00	96,00
Mokopane Layers Total			94,40	92,00	96,00
Polokwane Hatchery Hub	Broiler	Q1	94,50	93,00	96,00
		Q2	94,43	93,00	96,00
	Broiler Total		94,47	93,00	96,00
Polokwane Hatchery Hub Total			94,47	93,00	96,00
Tzaneen Breeder Farm	Breeder	Q1	95,25	93,00	97,00
		Q2	95,25	93,00	97,00
	Breeder Total		95,25	93,00	97,00
Tzaneen Breeder Farm Total			95,25	93,00	97,00
Grand Total			94,69	91,00	97,00

PIVOT 3 — Total Chick Output by Farm

Total Eggs							
Farm_Name	Farm Type	Graded 1st	Graded 2nd	Culled	Hatched	Avg Cull Rate %	Weeks
Bela-Bela Agri	Broiler	310600	6031	1969	318600	0,62	20
		310600	6031	1969	318600	0,62	20
Giyani Poultry	Broiler	304000	5519	2081	311600	0,67	19
		304000	5519	2081	311600	0,67	19
Lephalale Poultry Farm	Broiler	376200	6047	1953	384200	0,51	20
		376200	6047	1953	384200	0,51	20
Mokopane Layers	Layer	252320	3405	595	256320	0,23	20
		252320	3405	595	256320	0,23	20
Polokwane Hatchery Hub	Broiler	402100	7079	2421	411600	0,59	19
		402100	7079	2421	411600	0,59	19
Tzaneen Breeder Farm	Breeder	206000	3157	843	210000	0,40	20
		206000	3157	843	210000	0,40	20
Grand Total		1851220	31238	9862	1892320	0,50	118

PIVOT 4 — Monthly Production Summary

Row Labels	Total Eggs	Total Eggs	1st Avg Hatchab	1st Gra	Total Mor	Avg Fertility Rate	Biosec Test
Jan	446000	386950	91,47	378550	214	94,83	1
Fail	18500	15600	90,67	15200	13	93,00	1
Pass	427500	371350	91,50	363350	201	94,91	0
Feb	446000	382050	91,18	373650	256	93,92	1
Fail	18500	15200	90,29	14800	22	91,00	1
Pass	427500	366850	91,22	358850	234	94,04	0
Mar	427000	371270	91,43	363270	200	95,09	0
Pass	427000	371270	91,43	363270	200	95,09	0
Apr	446000	386550	91,46	378150	209	94,75	1
Fail	19000	15700	90,80	15300	14	91,00	1
Pass	427000	370850	91,49	362850	195	94,91	0
May	421000	365500	91,51	357600	193	94,87	0
Pass	421000	365500	91,51	357600	193	94,87	0
Grand Total	2186000	1892320	91,41	2E+06	1072	94,69	3

PIVOT 5 — QC Flag Summary

QCFlag	(All)	
Row Labels	QC Flag Test	Flag Rate %
Bela-Bela Agri	20	40,00%
Giyani Poultry	19	5,26%
Lephalale Poultry Farm	20	0,00%
Mokopane Layers	20	100,00%
Polokwane Hatchery Hub	19	5,26%
Tzaneen Breeder Farm	20	0,00%
Grand Total	118	

PIVOT 6— QC Flag Summary

Biosecurity_Check	(All)	
Row Labels	Biosec Test	Fail Rate %
Bela-Bela Agri	20	10,00
Giyani Poultry	19	5,26
Lephalale Poultry Farm	20	0,00
Mokopane Layers	20	0,00
Polokwane Hatchery Hub	19	0,00
Tzaneen Breeder Farm	20	0,00
Grand Total	118	

Dashboard KPIs at a Glance

Total Eggs Set 2,208,000	Chicks Hatched 1,911,520	Hatchability % 91.41%	Avg Fertility % 94.69%	1st Grade Chicks 1,872,810	Biosec Fails 3
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Key Insights

1. Best Performer

Tzaneen Breeder Farm achieved the highest average hatchability at 91.86%, consistent with optimal incubation conditions and strong fertility (95.25%).

2. Watch List

Mokopane Layers recorded the lowest average hatchability at 90.50%. Incubation temperature variance and egg handling procedures are worth investigating.

3. Biosecurity

Three biosecurity check failures were recorded across the period. Bela-Bela Agri (4 February) and Giyani Poultry (15 April) were repeat concerns, pointing to a need for stricter staff entry protocols.

4. Production Gaps

The Easter shutdown week (1 April 2022) caused a complete production loss across all six farms in one week. Better restart planning would reduce the impact of scheduled shutdowns.

5. Data Quality

30 weekly records (25% of the cleaned dataset) fell below the 91% hatchability threshold. Cross-checking these weeks against environmental log data (temperature, humidity) may reveal a common cause.

6. Feed Efficiency

Waterberg district farms (Lephalale, Bela-Bela, Mokopane) consumed more feed per hatched chick than farms in Capricorn and Mopani, warranting a feed conversion ratio analysis in the next phase.

7. Highest Cull Rate

Giyani Poultry recorded the highest cull rate overall (0.67%) and also the lowest fertility rate among the broiler farms — a possible shared root cause worth investigating (breeder flock age, incubation conditions, or handling).

8. Best Mortality Profile

Tzaneen Breeder Farm combined the lowest cull rate (0.40%) with the fewest mortalities (86 on 240,000 eggs set), consistent with its leading fertility and hatchability figures — a sign that a well-managed breeder flock drives gains across multiple metrics at once.

Recommendations

Address the Mokopane Hatchability Gap

- Review incubator calibration and temperature logs at Mokopane Layers against the better-performing farms.
- Audit egg handling and storage procedures between collection and setting.

Replicate Tzaneen's Breeder Performance

- Document what Tzaneen Breeder Farm does differently in flock management and incubation conditions.
- Use Tzaneen as the internal benchmark when training staff at lower-performing farms.

Tighten Biosecurity Compliance

- Enforce entry protocols at Bela-Bela Agri and Giyani Poultry, both repeat offenders.
- Consider a standard biosecurity checklist with sign-off before each incubation cycle.

Plan Around Scheduled Shutdowns

- Stagger flock scheduling around known shutdown periods (e.g. Easter) to avoid a full week of zero production across every farm at once.

Investigate Feed Efficiency in Waterberg

- Run a feed conversion ratio analysis comparing Waterberg farms to Capricorn and Mopani.
- Check feed quality and storage conditions specific to the Waterberg district.

Close Data Gaps

- Reconcile the one-row discrepancy between the documented cleaned row count (119) and the actual cleaned sheet (118 records).
- Convert text-stored numeric fields (Eggs_Fertile, Water_Consumed_L in places) to true numbers to avoid downstream calculation errors.
- Link environmental log data (temperature, humidity) directly to the 30 low-hatchability flags for root-cause analysis.

Dashboard



